Timothy Owolabi S.

AI Engineer — Backend & Platform Systems | Banking & Financial Services

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SUMMARY

Backend and platform engineer with 9 years building reliable, high-scale systems across financial services, agency banking, and e-commerce — including 2.5+ years in retail banking (BanqueMisr) and agency banking (3Line Card Management). Hands-on experience designing AI-driven, LLM-powered classification and categorization systems using structured prompting, and comfortable working across Python and Java/Kotlin stacks. Rome Business School Master's in Data Science Management complements a track record of leading engineering teams, owning systems end-to-end, and translating business processes into practical, production-grade software — well suited to identifying and delivering AI use cases in a banking environment.

RELEVANT AI & AUTOMATION EXPERIENCE

- Designed and built an AI-driven taxonomy classification system using structured JSON prompting against an LLM to automatically categorize keywords into a three-tier (C1/C2/C3) hierarchy, replacing manual categorization.
- Architected a self-hosted, on-premise keyword categorization and search service (FastAPI, local LLM via Ollama, PostgreSQL, Docker) — end-to-end ownership from concept through production.
- Applied AI inference to a German-language category-label simplification effort, using existing seed data to prompt-engineer consistent, human-readable outputs.
- Led replacement of a legacy Apache Mahout recommendation engine, evaluating modern implicit-feedback approaches (ALS-implicit, LightFM) and separating offline training from online serving — improving model load time by ~22%.

EXPERIENCE

Senior Software Engineer — Spreadgroup *Jun 2025 – Present*

- Extended Commercetools' checkout API with its advanced payment flow to integrate PayPal, Google Pay, and Apple Pay express payment methods, reducing checkout drop-off by an estimated 25% while supporting multiple payment rails in a single transaction flow.
- Designed custom API extensions hooking into Commercetools webhooks to enforce business logic and trigger downstream event processing in real time — comparable to ERP interface and data-exchange integration work.
- Re-architected the product-to-Coveo batch sync pipeline using streamed CSV processing and a tracking table across ~18 locales, cutting sync failures by an estimated 65% and improving data consistency at scale.
- Led the replacement of a legacy Apache Mahout recommendation engine, reducing model load time by an estimated 22% and improving serving performance.

Senior Software Engineer (Andela Network) — BanqueMisr — Retail Banking *Dec 2022 – May 2025*

- Migrated core retail banking services to GraalVM native image using the Spring buildpack, reducing microservice startup time and memory footprint.
- Refactored core logging libraries by customizing SLF4J to automatically redact sensitive data across banking services — a security and compliance-driven initiative directly relevant to regulated financial environments.
- Re-optimized the reporting service using CQRS and denormalized MongoDB collections to improve query performance.
- Increased test coverage for retail banking services, writing and reviewing integration and unit tests with Mockito, WireMock, and Castle Mock.

Senior Backend Engineer — Nomba *Sep 2022 – Nov 2023*

- Built core gRPC services and shared libraries (IAM, Storehouse) adopted by other teams, improving reusability and development speed.
- Developed an internal webhooks library for RESTful and SOAP callback delivery, with scheduled retry handling via Quartz — comparable to interface/data-exchange work between disparate systems.
- Integrated OpenID login via Auth0 and implemented authorization using CerBos.
- Reviewed existing codebases and modules to identify technical debt and code smells.

Software Architect & Technical Lead — Seamfix *Sep 2020 – Jul 2022*

- Technical Lead (9 months): led and mentored an 8-person engineering team on the Bioregistra product, defining roles, responsibilities, and delivery expectations within an agile process.
- Redefined CI/CD tooling and process, cutting deployment time by 25%.

- Reduced Kubernetes pod allocation by 20% through service and performance optimization, lowering operational costs.
- Senior Software Engineer (1 year): migrated the caching layer from Memcache to Redis, enabling synchronized multi-cluster caching, and converted key services from JEE to Spring microservices.

Software Engineer — 3Line Card Management Limited — Agency Banking *Apr 2019 – Aug 2020*

- Restructured core and shared services for the Freedom Network agency banking platform, growing customer engagement and revenue up to 10x within 3 months.
- Built the Cash Disbursement Application for NCTO using a Spring and Apache Kafka microservice architecture, supporting more than 3 million concurrent users.

TECHNICAL SKILLS

AI & Data: Python, FastAPI, LLM inference (Ollama), prompt engineering, structured-output prompting, PostgreSQL, data science fundamentals (Rome Business School MSc)

Core: Java (7/8/11/17), Kotlin, Spring Boot, Microservices, REST/GraphQL APIs, OAuth/SAML, Hibernate

Data & Messaging: PostgreSQL, MySQL, Oracle, MSSQL, MongoDB, Redis, Kafka, RabbitMQ

Infrastructure: AWS, Azure, Google Cloud, Docker, Kubernetes, GitLab CI/CD, Unix

Also familiar with: Golang, PHP, Node.js, React, Vue, Angular, Dropwizard, Digital Ocean, Heroku

EDUCATION

Masters in Data Science Management — Rome Business School (2024 – 2025)
Higher National Diploma in Computer Engineering — Yaba College of Technology, Nigeria (2015 – 2017)
Ordinary National Diploma in Computer Engineering — Yaba College of Technology, Nigeria (2012 – 2014)

LANGUAGES

English (Native) | Yoruba (Native)